PHYS1500J - Final RC

Zhao Guojiao

1. Rigid Body and Moment of Inertia

1.1 Some Concepts

Rigid Body

Definition: The distance between any two points of the rigid body remains constant (no compression, stretch, etc.).

Angular Quantities:

- angular displacement θ
- angular velocity ω
- angular acceleration ε

Comments:

- Note that θ, ω, and ε are vectors.

$$\mathrm{d}\vec{r} = \mathrm{d}\vec{\theta} \times \vec{r}$$

- Compare them with x, v, and a.

Moment of Inertia

General Calculation:
An object rotate along z-axis with the density function $\rho(x, y, z)$

$$I = \iiint_V (x^2 + y^2) \cdot \rho(x, y, z)\, dx\, dy\, dz$$

Practical Calculation Steps:

- Determine the axis. Find out how the body is symmetrical. Construct the equation based on the symmetry.
- Find out the $\mathrm{d}m$. Do integration.
- Substitute $\mathrm{d}m$ with m and other quantities.

Moment of Inertia

Definition:

$$I = \int r_\perp^2 \mathrm{d}m$$

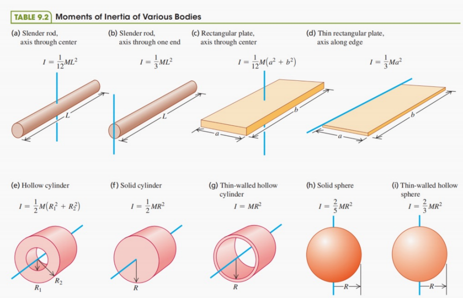
Relationship with Kinetic Energy:

$$K = \frac{1}{2} I \omega^2$$

Comments:

- I is the moment of inertia about the fixed axis of rotation A.
- Moment of inertia depends on the distribution of mass.

Moment of Inertia for Common Rigid Bodies



Parallel Axis Theorem

$$I_{A'} = I_A + mb^2$$

*I_A is the MoI with the axis passing through the center of mass.

Perpendicular Axis Theorem

$$I_Z = I_X + I_Y$$

*The rigid body is only on the plane of XoY.

1.2 Exercises

Two thin, uniform rods with mass m and length l are symmetrically connected to form a T-square ruler. Each part of the ruler is provided with a rotating shaft perpendicular to the ruler plane and the moment of inertia is I. Find I_{min} and I_{max}.

A uniform square thin plate has a mass of m and each side length of a. Take the rotating axis through center O on the plane of the plate, and find the moment of inertia of the plate with respect to the axis.

The semi-major axis length of an elliptical ring is a, the semi-minor axis length is b, and the mass is m (not necessarily homogeneous). The moment of inertia about the major axis is I_a, find the moment of inertia I_b about the minor axis.

2. Torque

2.1 Concepts

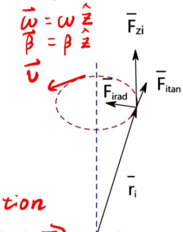
Definition

$$\overline{\tau} = \overline{r} \times \overline{F}$$

Rotation with Fixed Axis

The net force can be decomposed into three components: radial, tangential and along the z-axis $\bar{F}_i = \bar{F}_{i,rad} + \bar{F}_{i,tan} + \bar{F}_{i,z}$

$$\bar{F}_i = \bar{F}_i = \underbrace{\bar{F}_{i,rad}}_{\text{radial}} + \underbrace{\bar{F}_{i,tan}}_{\text{tangential}} + \underbrace{\bar{F}_{i,z}}_{\text{along the z-axis}}$$



the only component that contributes to rotation

if the motion is in a plane $\Rightarrow$ ω, β along z axis

Formula

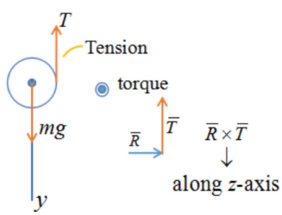
Formula:

$$\tau_z = I_z \varepsilon_z \quad \Rightarrow \quad f = ma$$

1.2 Two important examples

Thread Unwinding from Cylinder

free body diagram.
forces + torques

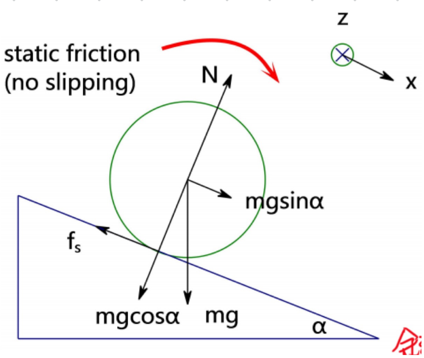


1. $mg - T = ma_{cm,y}$
2. $TR = I_{cm}\varepsilon_z = \frac{1}{2} mR^2 \varepsilon_z$
3. $a_{cm,y} = R\varepsilon_z$ (no slipping; follows from $v_{cm,y} = \omega_z R$)

$$a_{cm,y} = \frac{2}{3} g$$

$$T = \frac{1}{3} mg$$

Sphere Rolling Down an Incline (no slipping)



1. Explain the direction of static friction
2. Is static friction doing work?

1. imagine small angle α

   the reason why sphere rotate ⊗ this way.

2. translational kinetic energy

   rotational kinetic energy

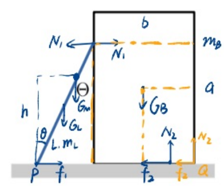
$$\begin{cases} ma_{cm,x} = mg\sin\alpha - f_s \\ f_s R = I_{cm}\varepsilon_z = \frac{2}{5}mR^2\varepsilon_z \\ a_{cm,x} = \varepsilon_z R \end{cases}$$

1.3 Exercise

Problem 3. (*15 points*) A man of mass m_M climbs up an unsecured ladder of length L and mass m_L. It is resting on the side of a large empty box of height a and width b, which has mass m_B. There is no friction between the ladder and the box.

(a) If the ladder makes an angle of θ degrees to the vertical side of the box, find the coefficient of friction between the box and the ground needed to prevent the box from sliding away from the ladder when the man stands at a height h above the ground.

(b) Assuming the friction is sufficient to prevent the box from sliding, find the maximum height the man can climb before the box begins to tip over.



The acceleration due to gravity g is known.

(*5 points*) A certain object with unknown shape is rotating about a fixed vertical axis passing through its center of mass. A scientist measures the part of the object farthest from the object's center of mass to be moving with speed v_1 at radius R_1. To measure the moment of inertia a scientist drops a stationary disk with $I = \frac{1}{2}M_d R_d^2$ on top of the unknown object. The disk is dropped such that its axis of symmetry (perpendicular to the disk's plane) is collinear with the axis about which the unknown object is rotating. The two objects hit and stick together and rotate as one object following the collision. With the addition of the disk the angular speed is found to decrease to 50% of its original value.

no use.

If the mass of the unknown object is equal to $\frac{3}{2}M_d$ and $R_d = R_1$, what is its moment of inertia?